

Yokohama City University - Data Analysis with R

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シラバス参照

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シラバス参照

科目名

Course name

Topics in Data Analysis

担当教員

Instructor

御器谷 裕樹

単位数

No of credits

2

期間・曜日・時限

Term/day/period

前期前半 水曜日 4 時限 YCUスクエア403教室

前期前半 水曜日 5 時限 YCUスクエア403教室

コース

・専攻名

Faculty

Data Analysis with R

学修到達

目標

Course

objective

In today's data-driven landscape, the ability to analyze and interpret complex datasets is essential for both rigorous academic research and strategic business decision-making. This course provides students with the foundational and advanced skills necessary to conduct robust data analysis using R, the industry-standard programming language for statistics and data science.

By the conclusion of this course, students will be proficient in the R ecosystem, including data manipulation, advanced regression modeling, and sophisticated data visualization. A key focus will be on reproducible research and reporting, ensuring students can present findings clearly for both scholarly publications and corporate environments. Through hands-on projects with real-world datasets, students will be prepared for careers in analytics, consulting, and research institutions.

授業概要・形態

Lecture Outline

and Class Format

This course provides a comprehensive foundation in statistical principles through the practical application of R. Students will master key topics in data analysis and visualization-skills that are indispensable in both academic research and professional business environments.

The curriculum begins with an introduction to the R ecosystem, specifically focusing on data wrangling-the reading, cleaning, and management of diverse datasets. Building on these technical skills, students will explore statistical inference and probability to solve complex, practical problems.

As effective communication of data is crucial, the course emphasizes data visualization. Students will learn to leverage R's powerful graphing libraries (such as ggplot2) to create impactful visual narratives.

Furthermore, the course covers essential statistical techniques, including hypothesis testing and regression analysis, while teaching students how to interpret intricate data structures.

The final module focuses on real-world integration. Students will apply their analytical toolkit to solve a real-world problem. This course equips students with the necessary expertise for careers in business analytics and research, where data-driven decision-making is paramount.

成績評価

方法

Grading

methods

Homework Assignments: 28%

Final Exam: 40%

Short Paper: 32%

This is an introductory-level course. Students who demonstrate an understanding of the material covered in the course and are able to independently master the core concepts will receive the highest evaluation.

前提科目

Prerequisite

courses

No prior knowledge of R programming is required. This course is designed for beginners. Comprehensive support will be provided. Therefore, students who are new to R programming or those who may not be confident in their English skills are highly encouraged to enroll.

事前事後学修

及び学修上・

履修申請上の

留意点

Pre- and

post-study

and points

to keep in mind

when studying

and applying

for courses

Students are required to bring their laptops to class and will engage in hands-on exercises, applying the methods covered in the lectures.

教科書・

参考図書等

Textbooks,

reference books,

etc.

1. Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). R for data science. O'Reilly Media, Inc.

2. Llaudet, E., & Imai, K. (2023). Data analysis for social science: A friendly and practical introduction. Princeton University Press.

Both English and Japanese versions of the materials are available, so I encourage those who are less confident in English to actively participate in this class as well.

教員研究室

Faculty

Laboratory

教員オフィス

アワー

Office

Hours

ホーム

ページ

Homepage

教員の

実務経験

について

Work

Experience

of Faculty

アクティブ

ラーニング

について

Active

Learning

[Type 1] "Lecture-centered active learning"

【類型1】「講義中心アクティブラーニング」

持続可能な

開発目標（SDGs）

17の目標との関連性

Relevance

to the 17

Sustainable

Development

Goals(SDGs)

SDGs

1. SDG 4: Quality Education

The course provides high-quality education by equipping students with essential data analysis skills that are in demand in today's data-driven economy. By teaching practical applications of R, statistical modeling, and data visualization, the course promotes inclusive, quality education and fosters lifelong learning opportunities.

2. SDG 9: Industry, Innovation, and Infrastructure

The course encourages innovation and supports infrastructure by fostering data analysis expertise, which is critical for developing solutions to real-world problems. The course also promotes the use of data-driven decision-making in industries and academia, contributing to innovation in these sectors.

授業計画

(週単位)

Course

plans

Class 1: Introduction to R

- Topics:

- o Introduction to the R environment
- o Installing and configuring R and RStudio
- o Basic R syntax, operators, and data types

- Learning Outcomes:

- o Students will be able to navigate the RStudio interface and use basic R commands.
- o Understand R as a programming language for statistical analysis.

- Activities:

- o Hands-on lab session: Basic R coding exercises (creating variables, basic arithmetic, and simple functions).

Class 2: Data Management, Data Structures, Functions, Lists and Loops

- Topics:

- o Data structures in R (vectors, lists, matrices, data frames)
- o Data manipulation using dplyr and tidyr
- o Writing custom functions in R
- o Looping constructs (for, while) and their use

- Learning Outcomes:

- o Understand and work with various data structures in R.
- o Manipulate and reshape datasets for analysis.
- o Write functions and automate repetitive tasks in R.

- Activities:

- o Data wrangling exercises in R using real-world datasets.

Class 3: Reading Data, Data Visualization with ggplot2

- Topics:

- o Importing and exporting data (CSV, Excel, etc.)
- o Introduction to basic plotting (histograms, scatterplots, etc.)
- o Introduction to the ggplot2 package for data visualization
- o The grammar of graphics: understanding the structure of ggplot2

- o Creating different types of plots (scatterplots, bar charts, boxplots, etc.)

- o Customizing plots (colors, themes, labels, and legends)

- o Faceting for multi-panel plots

- Learning Outcomes:

- o Read and manage data files in R.

- o Create and customize basic graphs for data visualization.

- o Understand the principles of ggplot2 and its grammar of graphics framework.

- o Create a variety of customized and informative visualizations using ggplot2.

- o Enhance data storytelling through effective visual presentation and customization.

- Activities:

- o Data import exercises and basic graph plotting.

- o Hands-on exercises to create different types of plots (scatterplots, bar charts, histograms) using ggplot2.

- o Customization exercises: modifying colors, labels, themes, and adding annotations to graphs.

- o Practice faceting to create multi-panel visualizations for comparative analysis.

Class 4: Estimating Causal Effects with Randomized Experiments

- Topics:

- o Introduction to causal inference and the counterfactual framework

- o Randomized Controlled Trials (RCTs): Project STAR case study

- o Individual vs. Average Causal Effects (ACE)

- o R skills: Relational operators (`==`, `%in%`, etc.), creating and subsetting variables

- Learning Outcomes:

- o Understand the fundamental problem of causal inference.

- o Explain how randomization allows for the estimation of causal effects.

- o Use R to subset data and calculate difference-in-means.

- Activities:

- o Hands-on lab: Analyzing the effect of small class sizes on student performance using relational operators and data subsetting.

Class 5: Inferring Population Characteristics via Survey Research

- Topics:

- o Principles of survey research and random sampling

- o Measuring public opinion: The Brexit Referendum case study

- o Descriptive statistics: Mean, median, and standard deviation

- o Visualizing distributions: Histograms and density plots in R

- Learning Outcomes:

- o Distinguish between sample statistics and population parameters.

- o Identify potential biases in survey research (non-response, sampling bias).

- o Generate and interpret frequency tables and univariate visualizations.

- Activities:

- o Data analysis exercise: Calculating Brexit support levels and visualizing the distribution of voter age and education using histograms.

Class 6: Predicting Outcomes Using Linear Regression

- Topics:

- o Introduction to the linear regression model (OLS)
- o Predictors, coefficients (intercept and slope), and residuals
- o Case study: Predicting GDP using night-time light emissions
- o Model fit: The Coefficient of Determination
- o Logarithmic transformations for skewed data
- Learning Outcomes:
 - o Understand the logic of "least squares" and how to fit a line to data.
 - o Interpret regression coefficients in a practical context.
 - o Assess model performance using R^2 .
- Activities:
 - o Hands-on exercise: Running a simple linear regression in R using the `lm()` function and interpreting the output.

Class 7: Estimating Causal Effects with Observational Data

- Topics:
 - o The challenge of confounding variables in non-experimental data
 - o Internal vs. External validity
 - o Introduction to Multiple Linear Regression
 - o Case study: The effect of Russian TV on Ukrainian voting behavior
- Learning Outcomes:
 - o Identify "omitted variable bias" and how it affects causal claims.
 - o Understand how multiple regression "controls" for confounders.
 - o Critically evaluate the validity of observational versus experimental studies.
- Activities:
 - o Regression lab: Comparing simple and multiple regression models in R to see how coefficients change when controls are added.

Class 8: Probability and Probability Distributions

- Topics:
 - o Basic axioms of probability and random variables
 - o Common distributions: Bernoulli and Normal distributions
 - o The Law of Large Numbers (LLN)
 - o The Central Limit Theorem (CLT)
- Learning Outcomes:
 - o Understand probability as the foundation of statistical inference.
 - o Recognize the properties of the Normal distribution (the "Bell Curve").
 - o Grasp how sample means behave as sample size increases.
- Activities:
 - o R Simulation: Conducting "virtual coin flips" and sampling exercises to visualize the Law of Large Numbers and the Central Limit Theorem.

Class 9: Quantifying Uncertainty: Confidence Intervals

- Topics:
 - o Sampling distributions and Standard Errors
 - o Constructing Confidence Intervals (CIs) for the sample mean
 - o CIs for the Difference-in-Means and regression coefficients

- o Visualizing uncertainty in R

- Learning Outcomes:

- o Define and calculate standard errors.

- o Interpret confidence intervals as a measure of estimation precision.

- o Predict outcomes with a specified margin of error.

- Activities:

- o Hands-on lab: Using R to calculate and plot confidence intervals for political and economic indicators.

Class 10: Quantifying Uncertainty: Hypothesis Testing

- Topics:

- o The logic of Null Hypothesis Significance Testing (NHST)

- o p-values and statistical significance

- o t-tests for comparing means

- o Statistical vs. Scientific (practical) significance

- Learning Outcomes:

- o Understand what a p-value actually represents (and what it doesn't).

- o Perform hypothesis tests for regression coefficients.

- o Avoid common pitfalls in the interpretation of "significance."

- Activities:

- o Exercise: Conducting t-tests and regression significance tests in R. Practice "writing up" the results of a hypothesis test in a scholarly format.

Class 11: Comprehensive Review and Practical Workshop

- Topics:

- o Review of the end-to-end data analysis workflow (from data cleaning to inference).

- o Summary of key concepts: Causality, Regression, and Uncertainty.

- o Common R coding pitfalls and debugging strategies.

- o Preparation for the Final Exam and the Short Paper.

- Learning Outcomes:

- o Synthesize technical R skills with theoretical statistical frameworks.

- o Determine the appropriate analytical method (e.g., difference-in-means vs. multiple regression) based on a given research question and data type.

- o Independently troubleshoot common errors in R scripts.

- o Translate statistical outputs into clear, substantive social science findings.

- Activities:

- o "Mock Analysis" Session: A guided, comprehensive exercise using a new dataset. Students will perform data importing, cleaning, visualization (ggplot2), and regression analysis in a single workflow.

- o Interactive Coding Quiz: A series of short exercises designed to test understanding of R syntax and statistical interpretation.

Class 12: Real-World Data Analysis Using V-Dem Dataset

- Topics:

- o Introduction to the V-Dem (Varieties of Democracy) dataset

- o Overview of key political and economic indicators in V-Dem

- o Analyzing economic trends across countries using statistical techniques

- Learning Outcomes:

- o Gain experience working with large, real-world datasets, specifically economic and political data.
- o Perform data manipulation, summary statistics, and visualizations using the V-Dem dataset.
- o Interpret key economic and political indicators to draw insights about global trends in democracy and governance.

- Activities:

- o Hands-on exercise analyzing the V-Dem dataset: students will explore variables related to economic factors, political participation, and democracy levels across various countries and regions.
- o Students will create summary statistics and visualizations (e.g., trends over time, country comparisons).
- o Discussion on the significance of findings in the context of global economic and political developments.

Class 13: Project Development and Consultation

- Topics:

- o Finalizing research questions and hypothesis formulation
- o Data cleaning and preprocessing for specific research topics
- o Selecting appropriate statistical models (e.g., Multiple Regression, RCT analysis)
- o Instructor consultation and peer feedback

- Learning Outcomes:

- o Develop a concrete analytical roadmap for the final project.
- o Troubleshoot specific coding challenges related to individual/group datasets.
- o Integrate statistical theory with practical R implementation to address research questions.

- Activities:

- o In-class Workshop: Students work on their data analysis scripts with real-time instructor support.
- o Consultation Sessions: Brief one-on-one or group meetings with the instructor to review research designs and preliminary findings.
- o Note: While the project may be completed in groups, each student will still be responsible for their individual Short Paper submission.

Class 14: Final Project Presentations

- Topics:

- o Presenting data-driven insights to a professional audience
- o Effective communication of statistical results and visualizations
- o Peer review and constructive academic discussion

- Learning Outcomes:

- o Effectively communicate complex data findings using ggplot2 visualizations.
- o Demonstrate the ability to interpret and defend statistical choices (e.g., model fit, variable selection).
- o Provide and receive constructive feedback on analytical methodologies.

- Activities:

- o Final Presentations: Students/groups present their research findings, focusing on the problem statement, methodology, and key insights.
- o Q&A Sessions: Active participation in discussing peers' work, simulating a professional research environment.

Class 15: Final Exam

備考

Remarks